GLOBAL PATIENT RECORDS · 2015–2024

Cancer Data Analysis

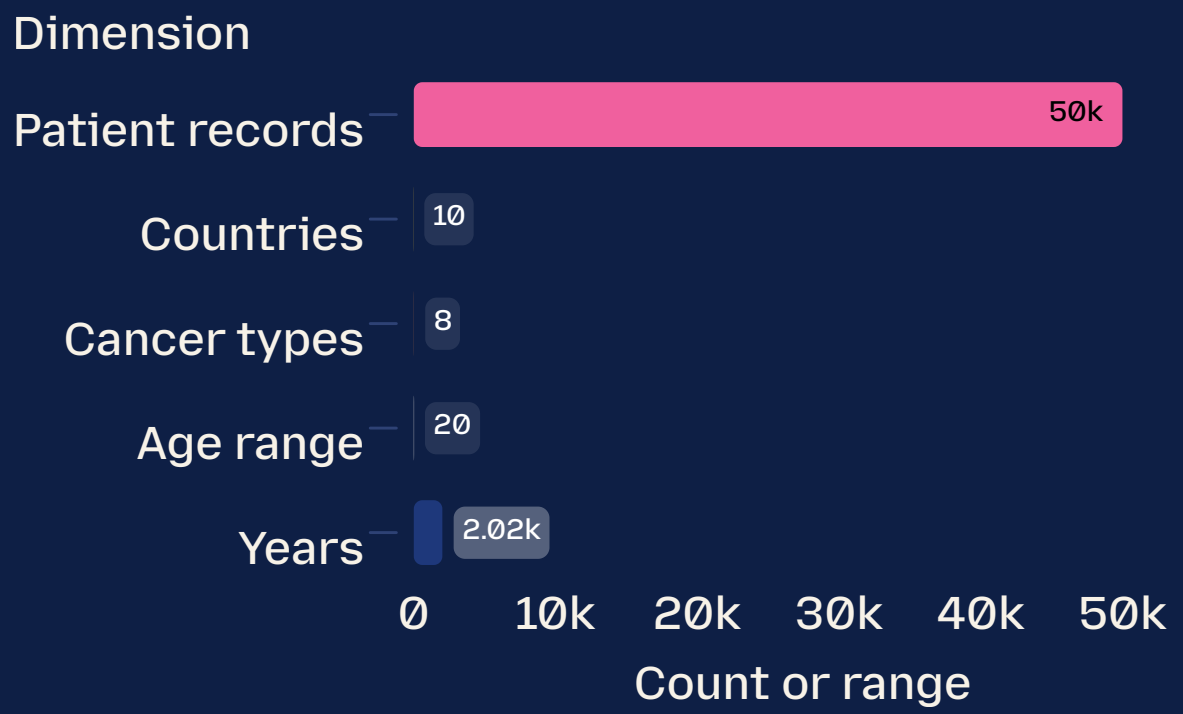
What 50,000 global patient records reveal about severity, early diagnosis, treatment cost, and survival.



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The dataset is broad, balanced, and unusually uniform

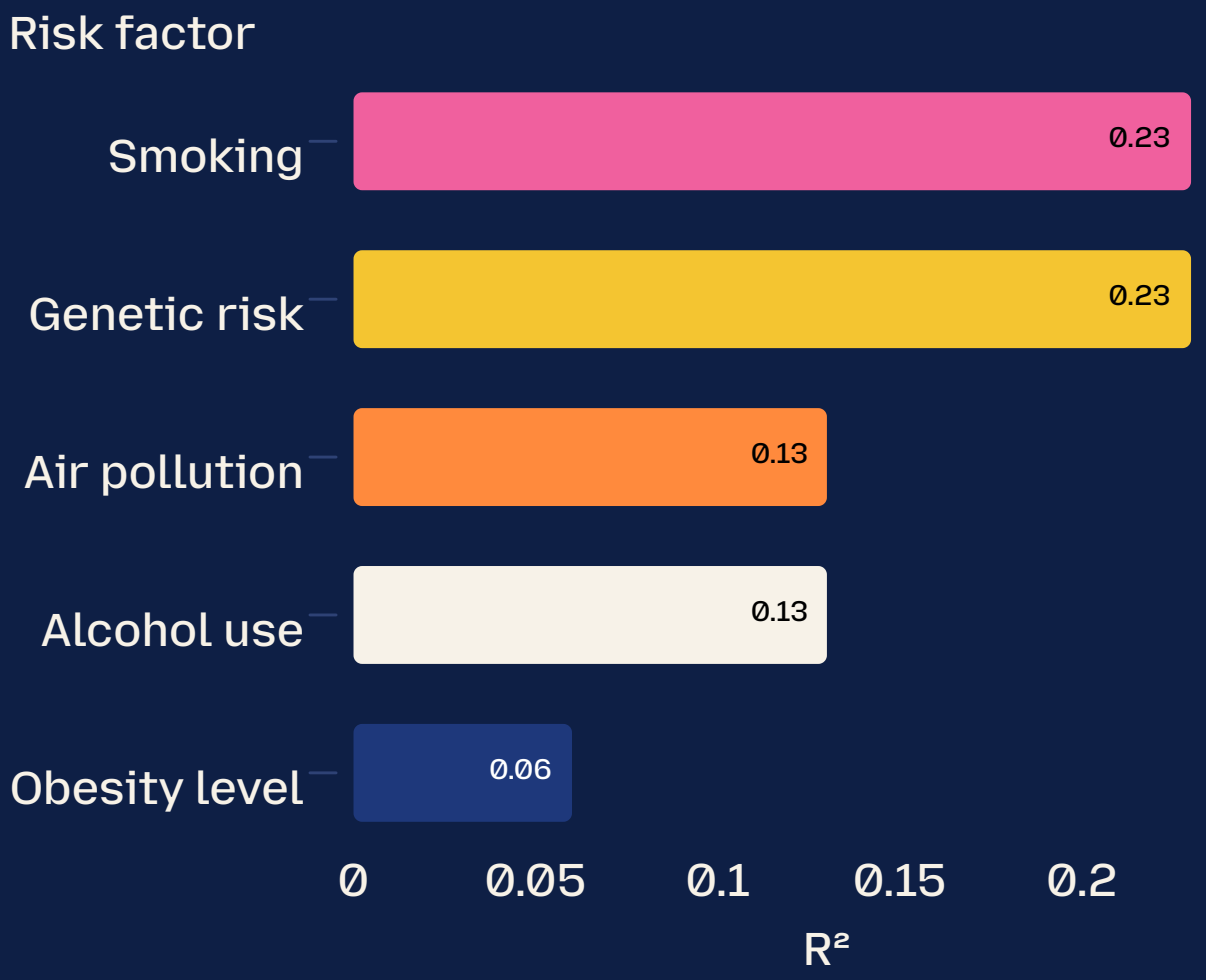
- 50,000 records (2015–2024), 10 countries, 8 cancer types
- Age 20–89; gender, country, cancer type, and stage near-evenly represented
- No missing values or duplicate rows reported



Age range: 20–89 · Years: 2015–2024

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Smoking and genetic risk lead the severity signal



INDIVIDUAL REGRESSIONS

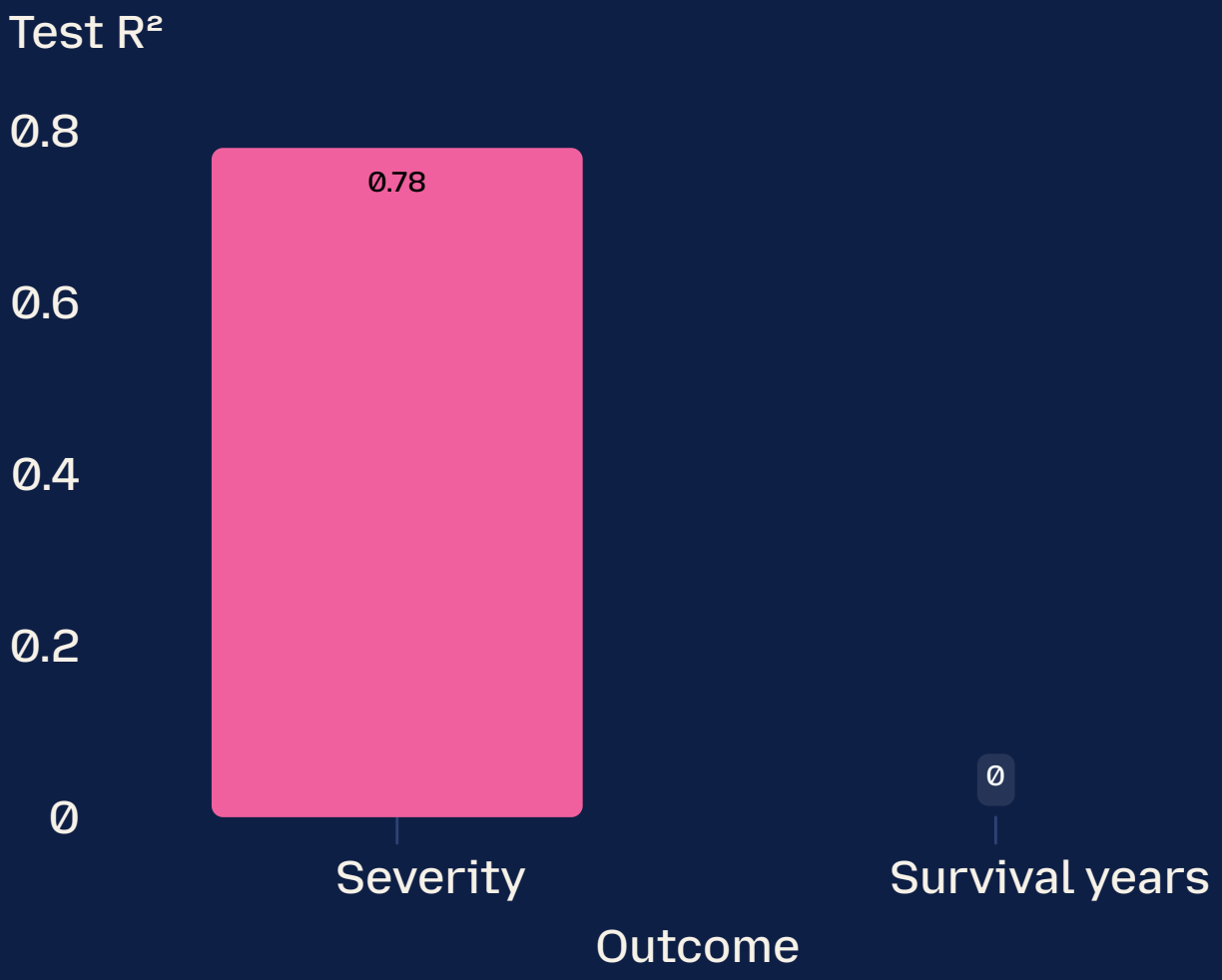
≈23%

Smoking and genetic risk each explain about 23% of the variation in severity.

Air pollution and alcohol use follow at about 13% each; obesity is weaker at 6%. All five risk factors show a positive relationship with severity.

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Severity is predictable; survival is not



0.78

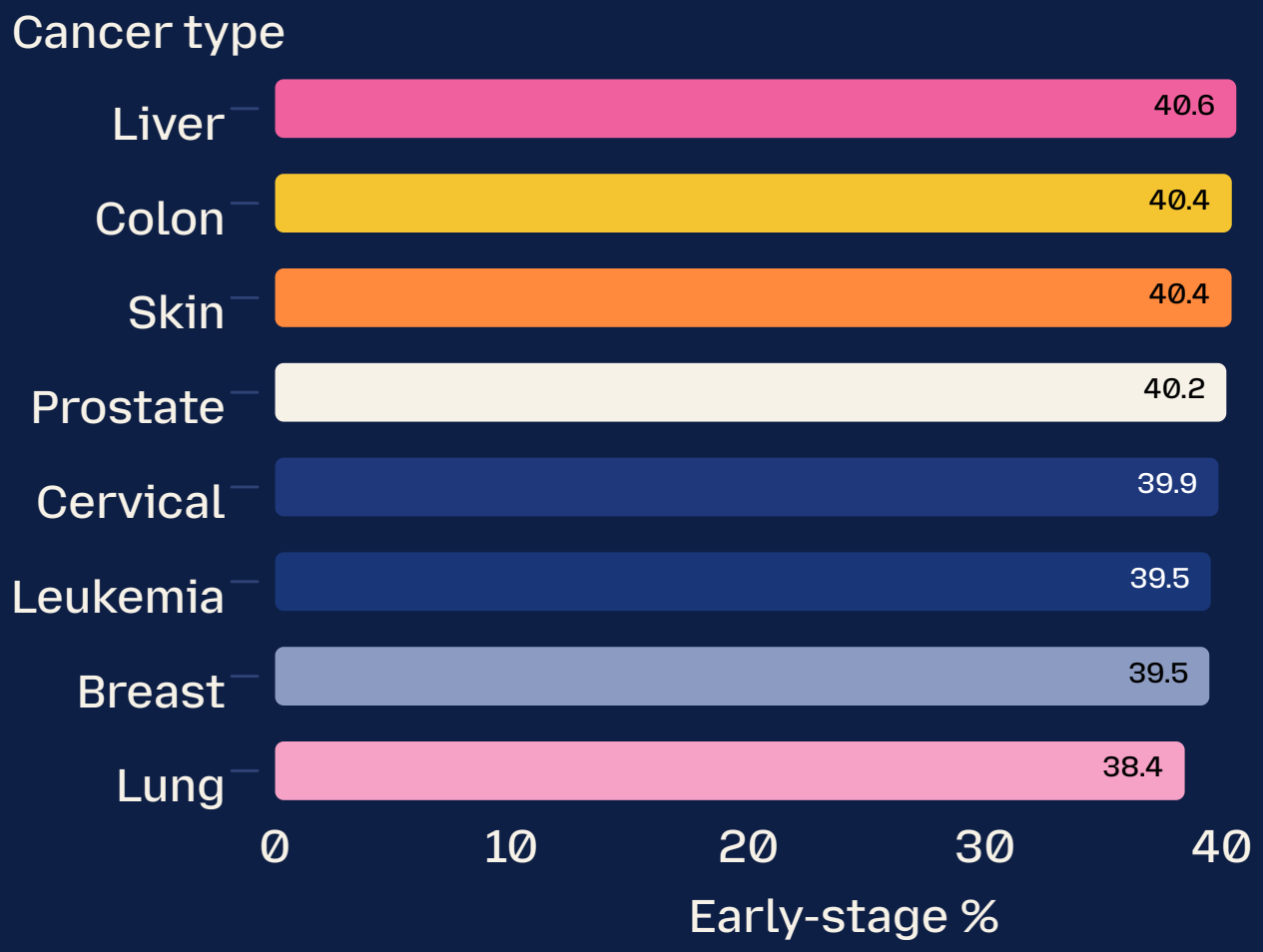
Random Forest test R² for severity

Severity correlations are strongest for smoking (r = 0.484) and genetic risk (r = 0.479).

A tuned survival model reached test R² ≈ 0.00, showing the available features explain almost none of survival variation.

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Early diagnosis is stuck near 40%



STAGE 0 + STAGE I

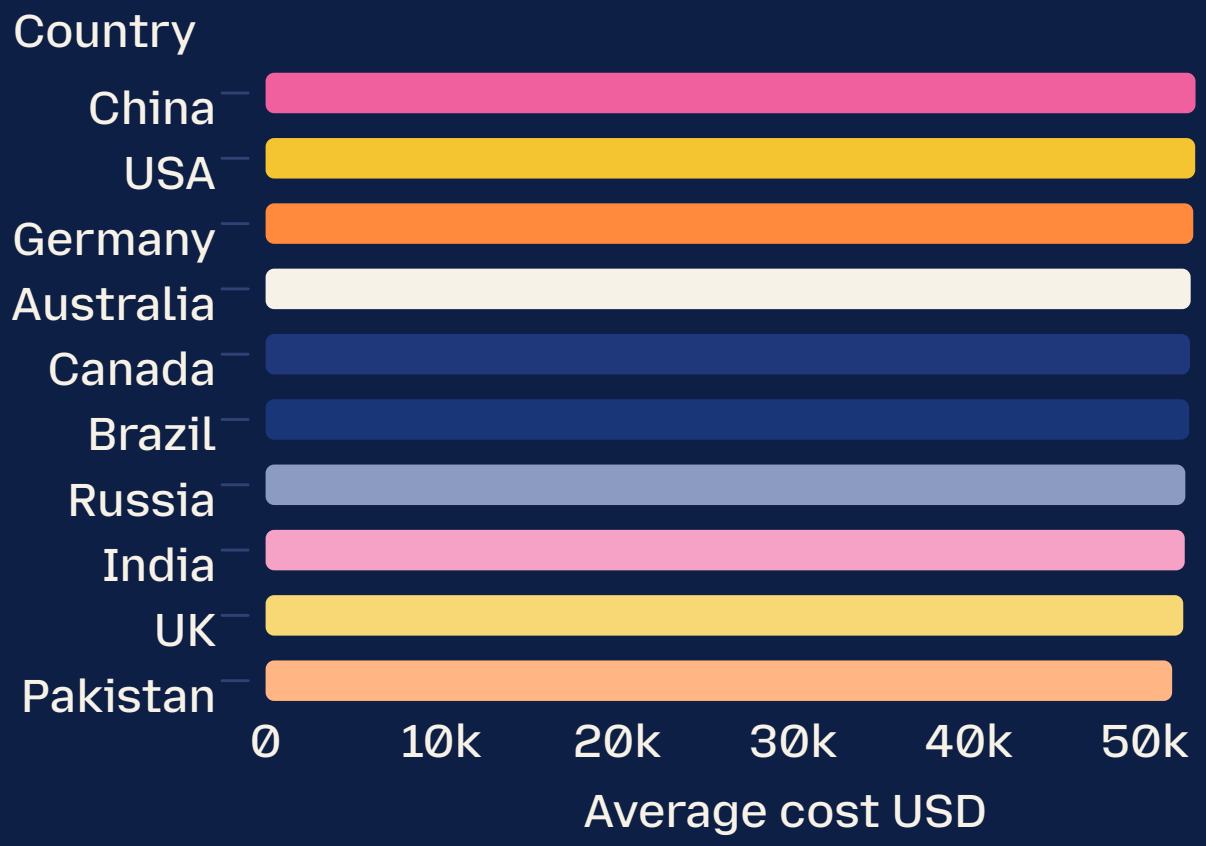
38–41%

The gap is only about two percentage points.

Liver is highest at 40.61%; lung is lowest at 38.43%. Lung remains the clearest screening opportunity.

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Treatment cost is nearly uniform across countries



$52,467

Average treatment cost

Range in the dataset: $5,000–$100,000.

Country averages span only about $1,300 from highest to lowest. Gender and age-group differences are similarly small.

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COST × SURVIVAL

Higher cost does not predict longer survival

r = −0.0004

Pearson correlation

p = 0.924

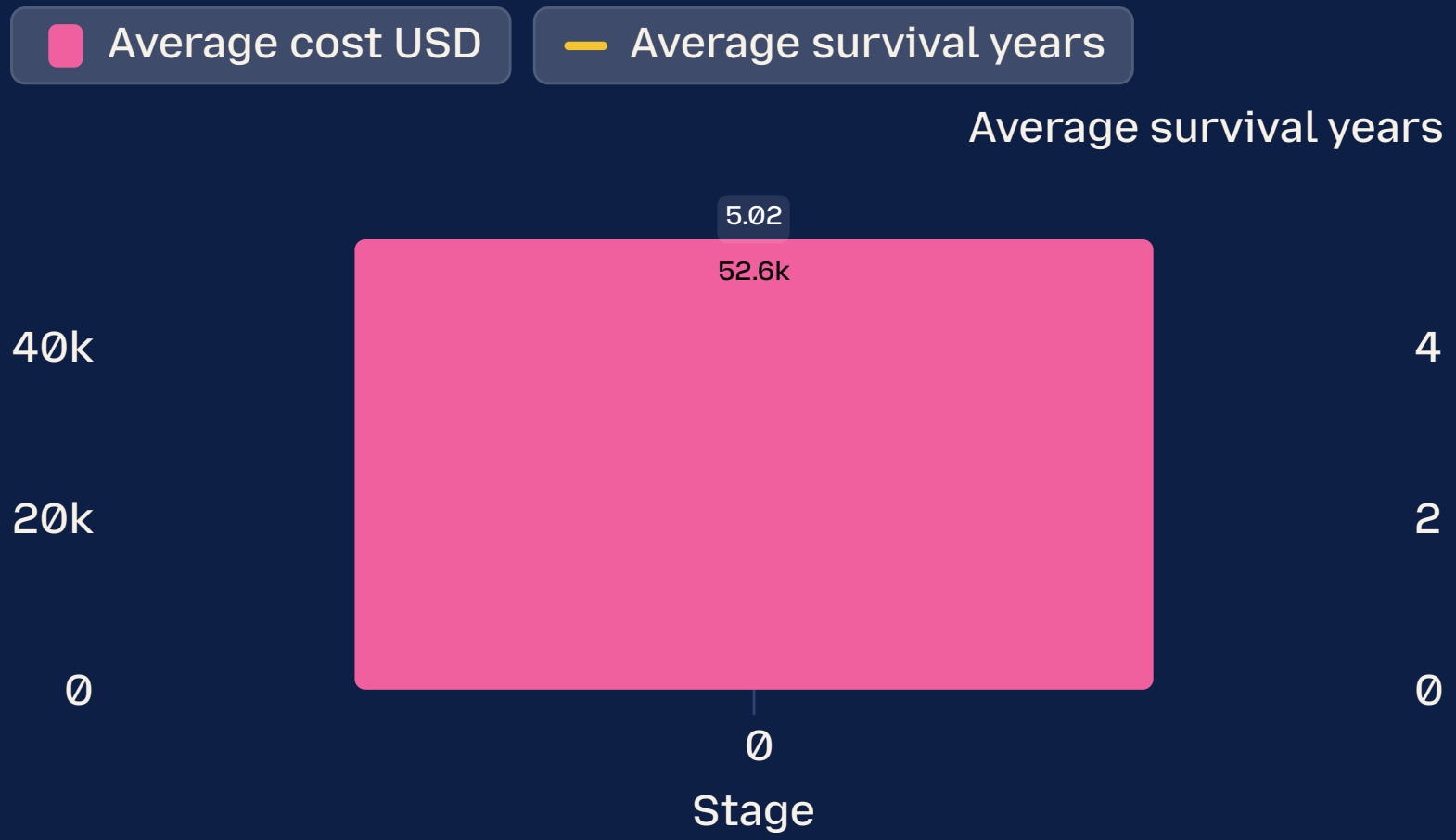
Association test

The analysis fails to reject the null hypothesis of no association.

In this dataset, spending more is not associated with surviving longer. Spearman correlation is also −0.0004, p = 0.921.

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Cancer stage does not separate cost or survival



STAGES 0–IV

Both averages are almost flat.

Average cost stays between $52,083 and $52,708. Average survival stays between 4.97 and 5.04 years.

Kruskal–Wallis: cost p = 0.425; survival p = 0.603.

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INTERACTION TEST

Genetic risk does not amplify smoking’s effect

The two factors appear to act independently on severity in this dataset.

−0.000228

Smoking × genetic-risk interaction coefficient

95% confidence interval: [−0.00115, 0.00069]

p = 0.628

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What the analysis supports—and what it cannot



Prioritize smoking cessation, air-pollution reduction, alcohol-risk programs, and obesity prevention.



Strengthen lung cancer screening to improve early detection.



Add treatment type, comorbidities, and follow-up dates before drawing conclusions about survival.



Interpret the uniform distributions and flat survival relationships cautiously: the report notes the dataset may be simulated; the severity model may overfit.

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